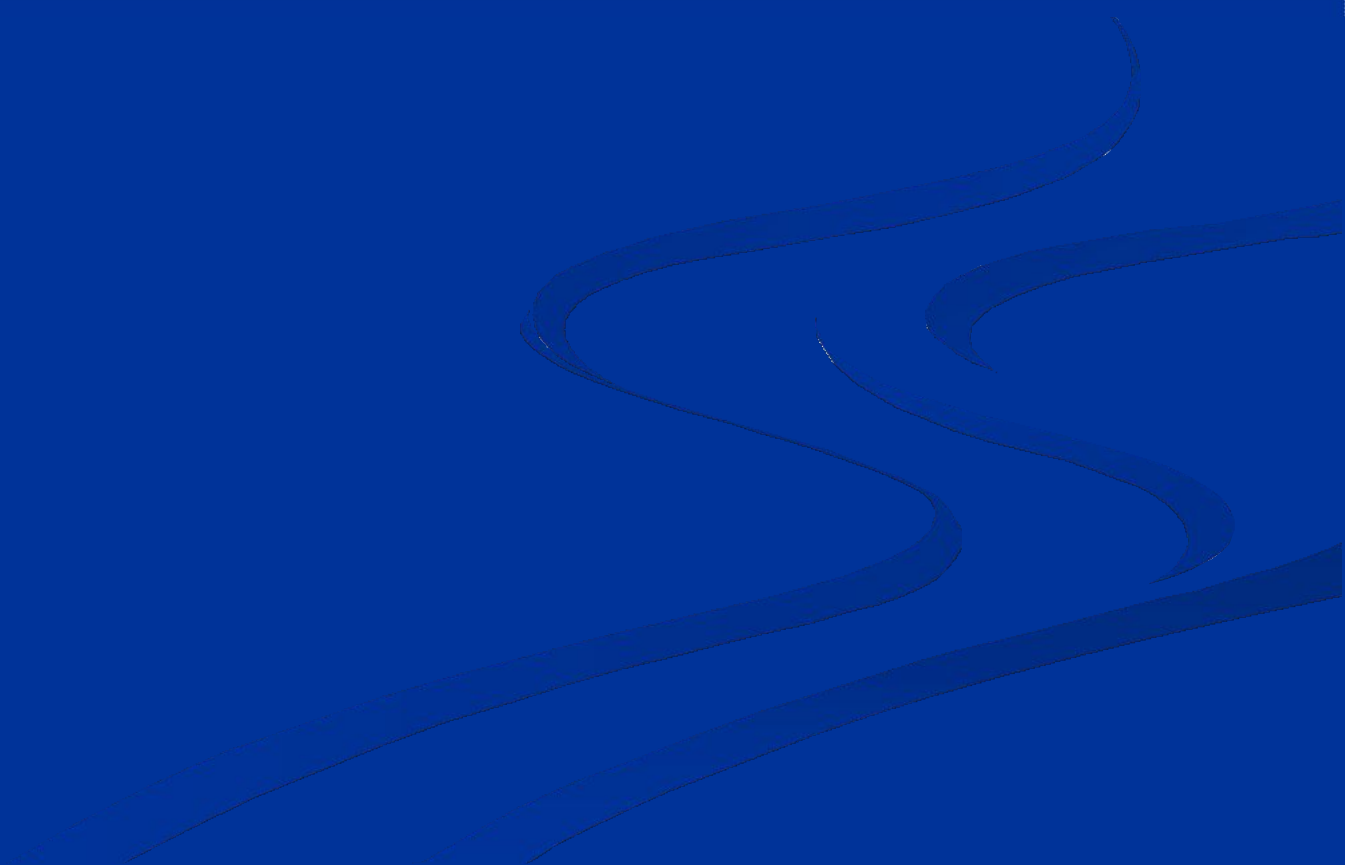


Chemistry of Nucleotides



Learning Objectives

1. Nitrogenous bases
2. Nucleosides
3. Nucleotides

Nucleotides are low molecular weight,
nitrogenous compounds

They are essential for life

They perform some vital functions
in all living organisms

Nucleotides act as:

Monomers of nucleic acids
High-energy compounds

Second messengers

Coenzymes

Metabolic intermediates

Nucleotides as monomers

Nucleotides are the monomeric units of nucleic acids

Ribonucleotides contain the sugar, ribose

Deoxyribonucleotides contain the sugar, deoxyribose

Nucleotides as high-energy compounds

Some nucleotides are high-energy compounds

An example is adenosine triphosphate
(ATP)

ATP is called the energy currency of cells

It is formed in exergonic reactions, and is used in endergonic reactions

Nucleotides as second messengers

Some nucleotides control biochemical and physiological processes by acting as second messengers for hormones

Examples are cyclic adenosine mono-phosphate (cAMP) and cyclic guanosine monophosphate (cGMP)

Nucleotides as coenzymes

Many nucleotides act as coenzymes:

Flavin adenine dinucleotide (FAD)

Nicotinamide adenine dinucleotide (NAD)

Nicotinamide adenine dinucleotide phosphate (NADP)

Nucleotides as metabolic intermediates

Some nucleotides act as intermediates in metabolic pathways:

Uridine diphosphate glucose

Uridine diphosphate glucuronic acid

Cytidine diphosphate diacylglycerol

Chemistry

Nucleotides are
made up of:

```
graph TD; A[Nucleotides are made up of:] --- B[A nitrogenous base]; A --- C[A sugar]; A --- D[A phosphate];
```

A
nitrogenous
base

A
sugar

A
phosphate

The nitrogenous base combines with sugar to form a nucleoside:



The nucleoside combines with phosphate to form a nucleotide:



Nitrogenous bases

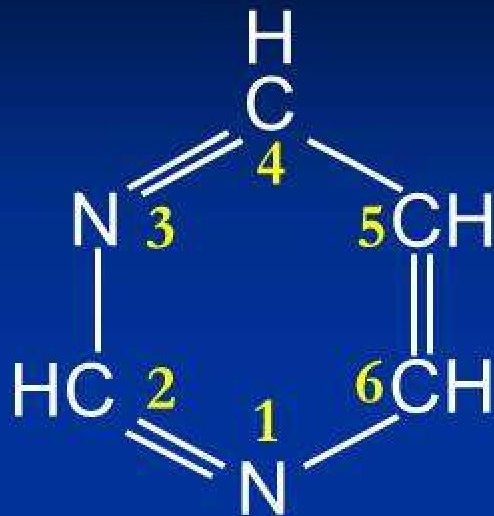
Two types of nitrogenous bases are found in nucleotides



Pyrimidine



Purine



Pyrimidine



Purine

Numbering of carbon and nitrogen atoms

The major
pyrimidine bases
in nucleotides are:

Cytosine

Uracil

Thymine

The major
purine bases
in nucleotides are:

Adenine

Guanine



Cytosine
(2-Oxy-4-amino-
pyrimidine)



Uracil
(2,4-Dioxy-
pyrimidine)



Thymine
(2,4-Dioxy-5-
methylpyrimidine)

Pyrimidine bases



Adenine
(6-Aminopurine)



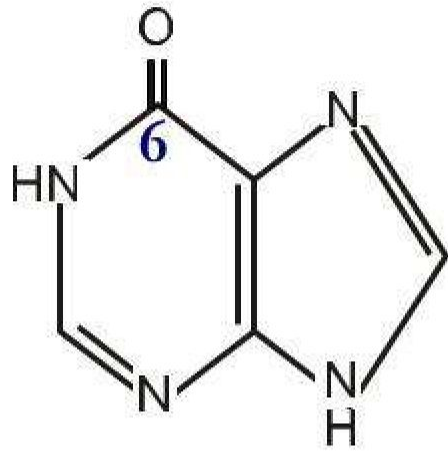
Guanine
(2-Amino-6-oxypurine)

Purine bases

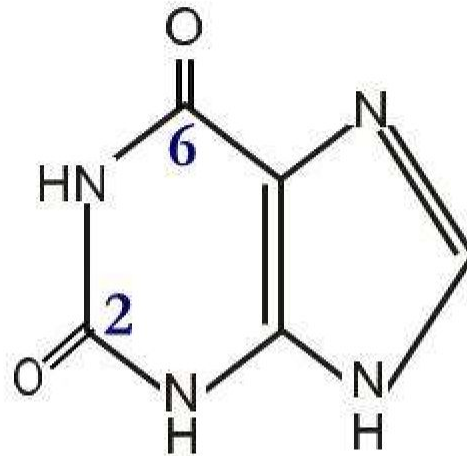
Hypoxanthine and xanthine, are metabolites of adenine and guanine

These are ultimately converted into uric acid

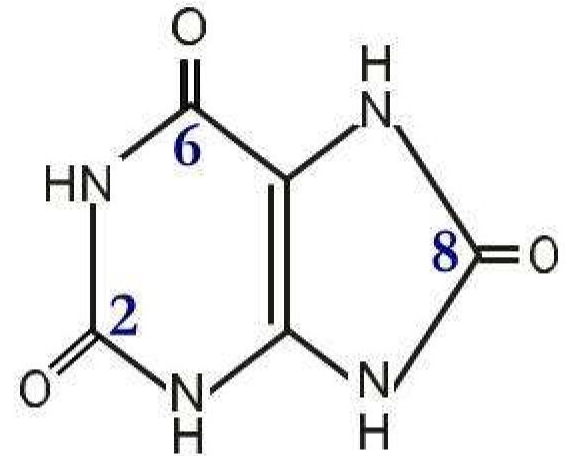
Uric acid is the end product of purine catabolism



Hypoxanthine
(6-Oxypurine)



Xanthine
(2,6-Dioxypurine)



Uric acid
(2,6,8-Trioxypurine)

Metabolites of purines

Some unusual bases found in nucleic acids, specially

RNA:

N⁶-Methyladenine

N⁶-Dimethyladenine

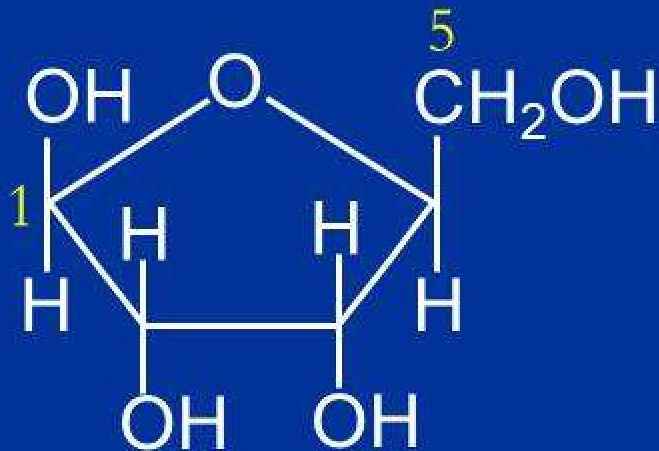
N⁷-Methylguanine

5-Methylcytosine

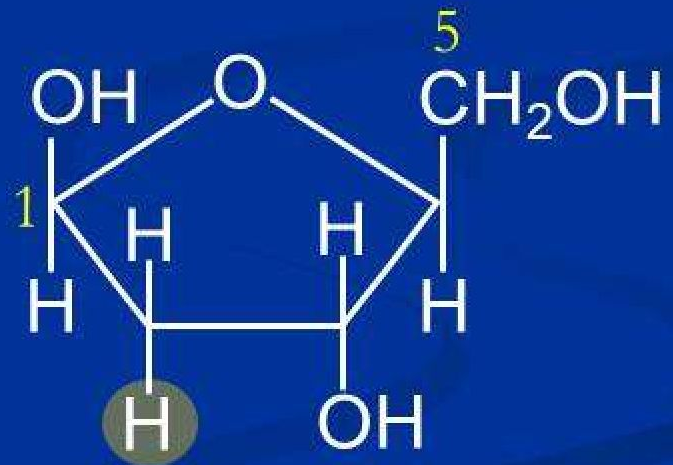
5-Hydroxymethylcytosines

Sugars

Two pentose sugars are found
in nucleotides



Ribose



Deoxyribose

Nucleosides

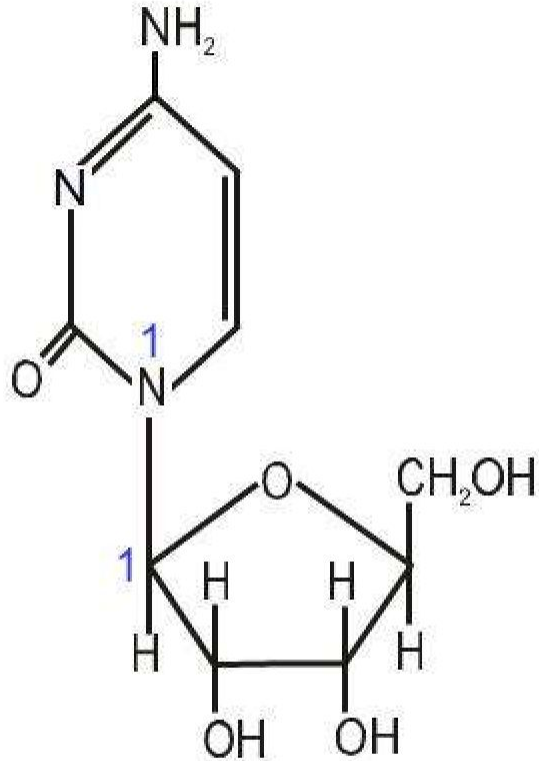
The sugar is linked to a base to form a nucleoside

A carbon-nitrogen bond is formed between:

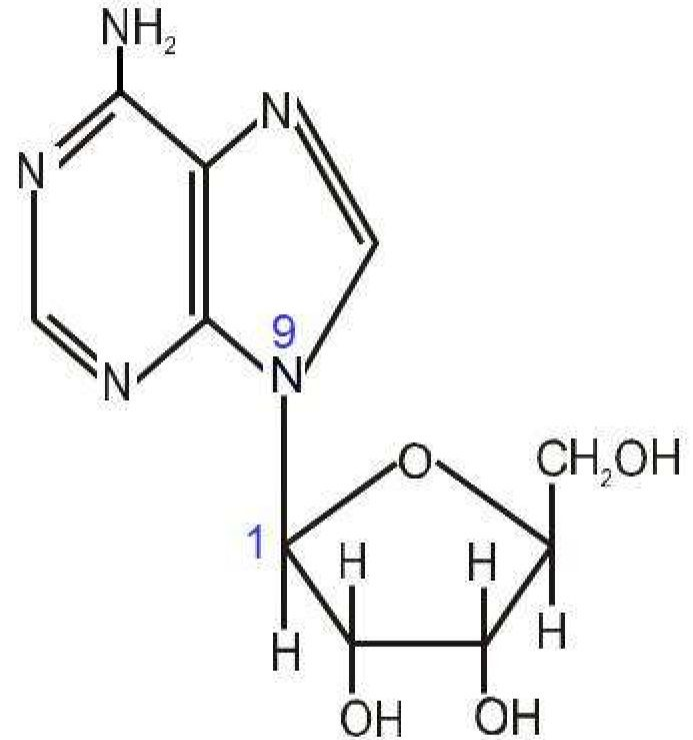
Carbon atom 1 of
sugar and nitrogen
atom 1 of the
pyrimidine base

Carbon atom 1 of
sugar and nitrogen
atom 9 of the
purine base

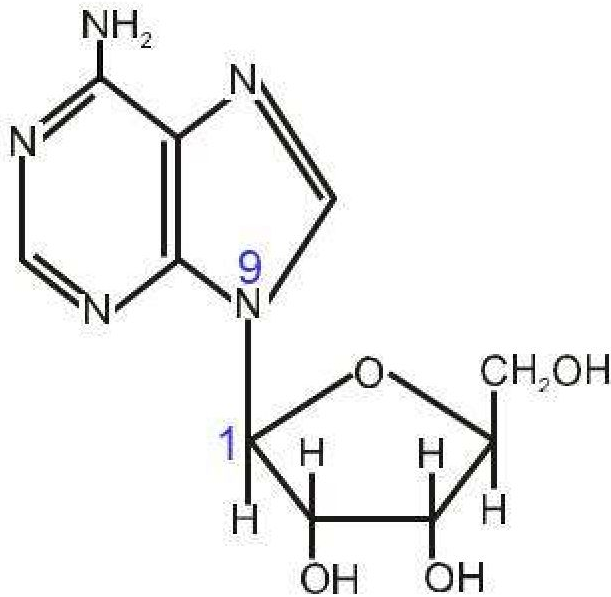
Nucleosides



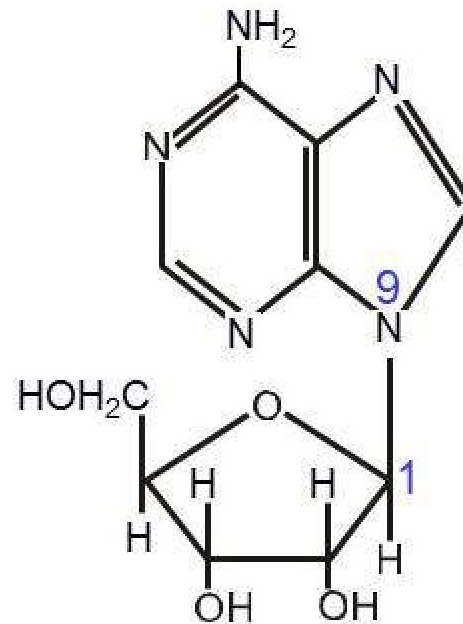
Cytidine
(cytosine + ribose)



Adenosine
(adenine + ribose)



anti conformation
(base and sugar are on
opposite sides of the
C–N bond)



syn conformation
(base and sugar are on
same side of the
C–N bond)

The *anti* conformation is more common

Nucleosides containing ribose are:

Cytidine (cytosine + ribose)

Uridine (uracil + ribose)

Adenosine (adenine + ribose)

Guanosine (guanine + ribose)

Nucleosides containing deoxyribose are:

Deoxycytidine (cytosine+deoxyribose)

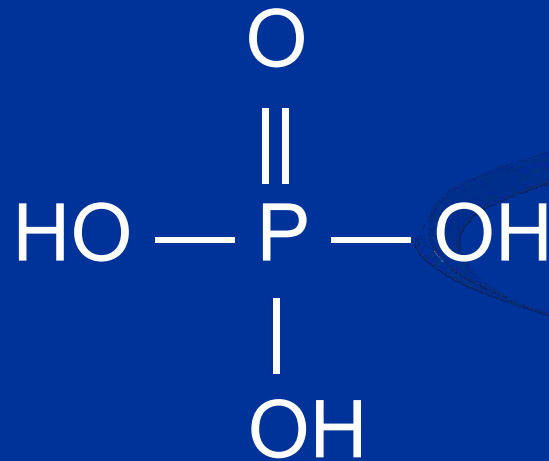
Deoxythymidine (thymine+deoxyribose)

Deoxyadenosine (adenine+deoxyribose)

Deoxyguanosine (guanine+deoxyribose)

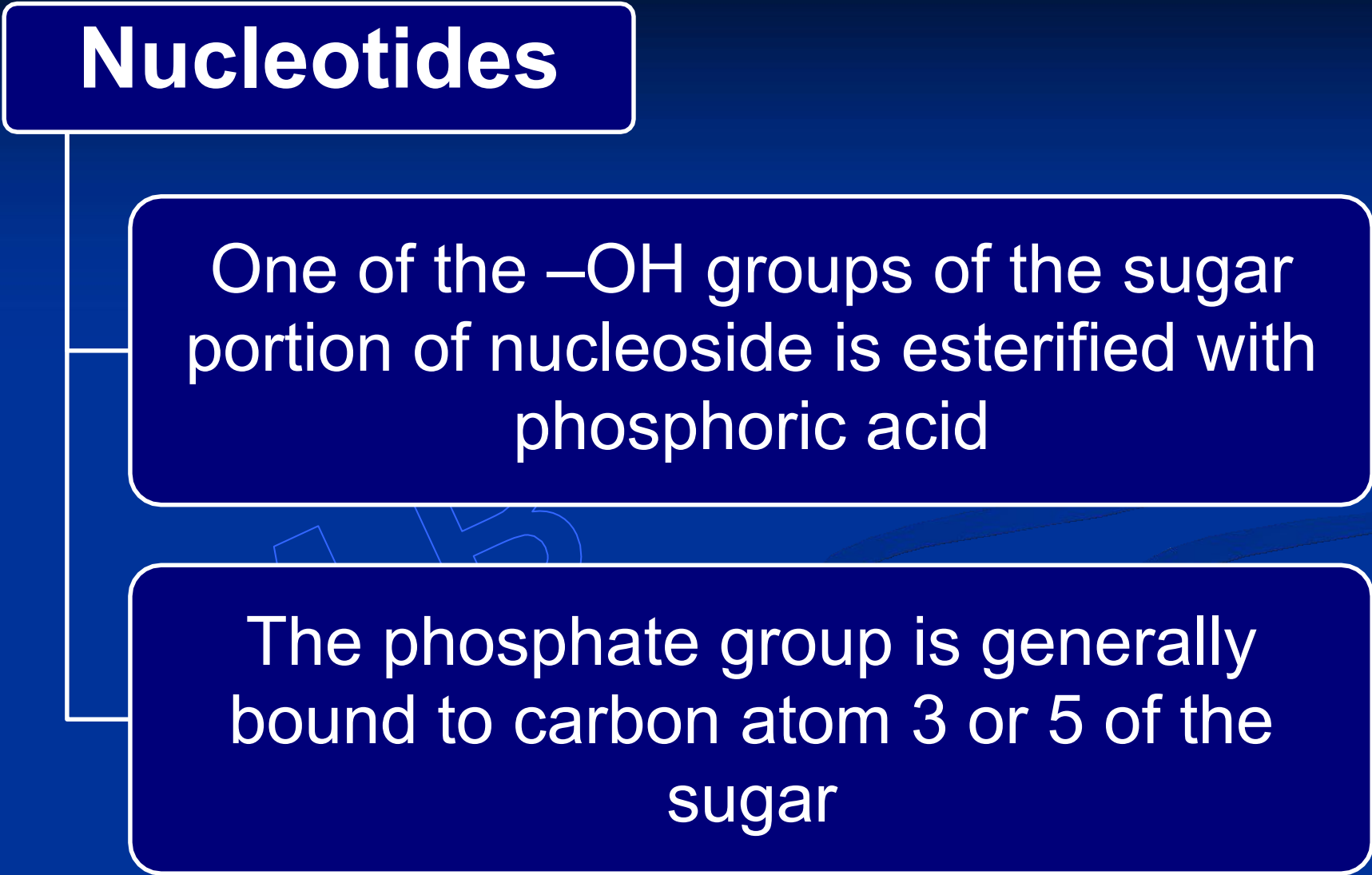
Phosphoric acid

The nucleosides combine with phosphoric acid (H_3PO_4) to form nucleotides



Phosphoric acid

Nucleotides



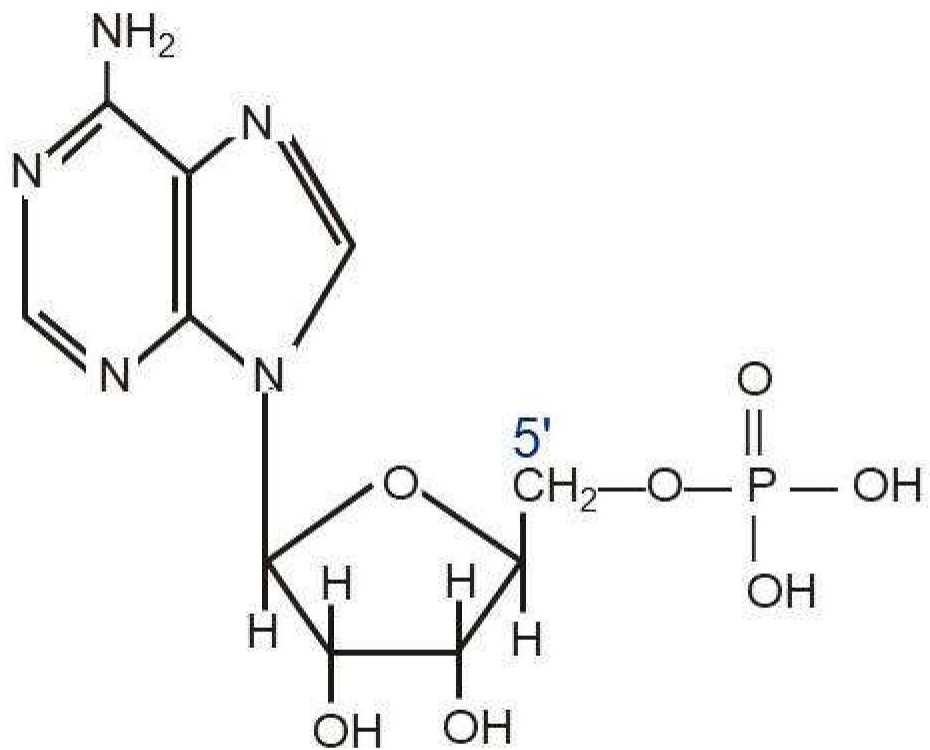
One of the -OH groups of the sugar portion of nucleoside is esterified with phosphoric acid

The phosphate group is generally bound to carbon atom 3 or 5 of the sugar

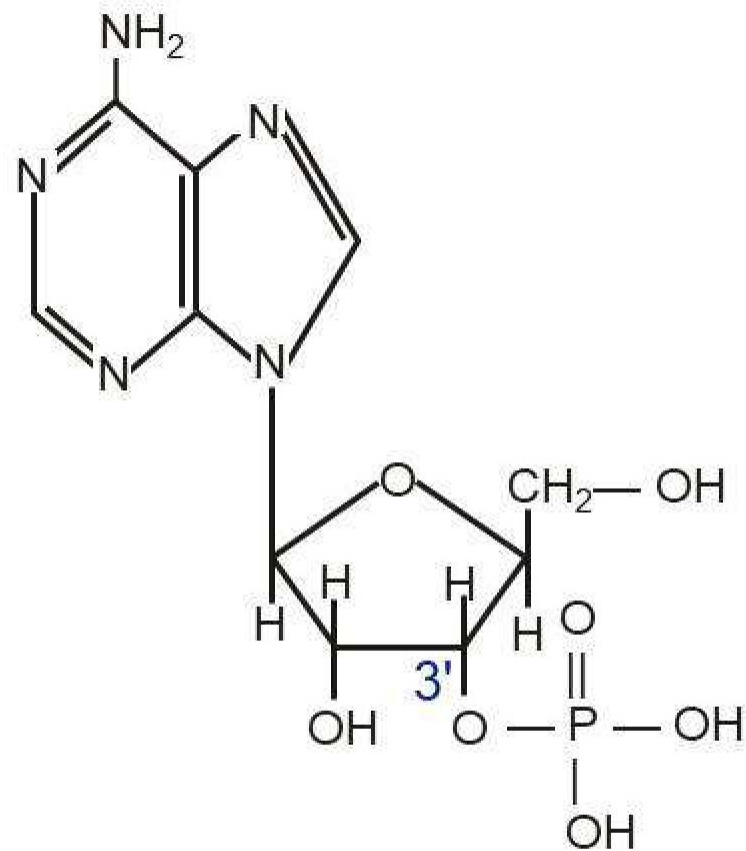
Numbering of atoms

To avoid confusion between numbering of various atoms in the base and the carbon atoms of the sugar, those of the latter are numbered 1', 2', 3' etc

The prime mark on the numeral shows that it is carbon atom of the sugar



Adenosine-5'-monophosphate



Adenosine-3'-monophosphate

Abbreviations

A single-letter abbreviation is commonly used for ribonucleosides

Adenosine, guanosine, cytidine and uridine are shown as A, G, C and U respectively

Deoxyadenosine, deoxyguanosine, deoxycytidine and deoxythymidine are shown as dA, dG, dC and dT respectively

A single phosphate group, attached to the sugar portion of the nucleotide, is represented by the letters MP

Nucleotide	Abbreviation
Adenosine monophosphate	AMP
Guanosine monophosphate	GMP
Cytidine monophosphate	CMP
Uridine monophosphate	UMP

These compounds are ribonucleotides

The letter 'd' precedes the abbreviations in deoxyribonucleotides

Nucleotide	Abbreviation
Deoxyadenosine monophosphate	dAMP
Deoxyguanosine monophosphate	dGMP
Deoxycytidine monophosphate	dCMP
Deoxythymidine monophosphate	dTMP

When the phosphate group is attached to carbon atom 5 of the sugar, its position is not indicated

Thus, AMP means adenosine-5'-monophosphate and soon

When phosphate is attached to a carbon atom other than 5, its position is indicated by a numeral

For example, A-3'-MP means adenosine- 3'-monophosphate

More than one groups are
phosphate present in some
nucleotides

First phosphate group is bonded to an -OH
group of the sugar by an ester linkage

Subsequent phosphates are attached to
preceding ones by acid anhydride bonds

When two phosphate groups are present in a nucleotide, they are denoted by the letters DP (diphosphate)

Three phosphate groups are shown by the letters TP (triphosphate)

Nucleotide	Abbreviation
Adenosine monophosphate	AMP
Adenosine diphosphate	ADP
Adenosine triphosphate	ATP

Some important nucleotides

Many nucleotides function as free nucleotides which include:

- Adenine derivatives
- Guanine derivatives
- Hypoxanthine derivatives
- Cytosine derivatives
- Uracil derivatives
- Vitamin derivatives
- Synthetic derivatives

Adenine derivatives

```
graph TD; A[Adenine derivatives] --> B[AMP, ADP and ATP are the nucleotides containing adenine]; A --> C[ADP and ATP are described as high-energy phosphates];
```

AMP, ADP and ATP are the nucleotides containing adenine

ADP and ATP are described as high-energy phosphates

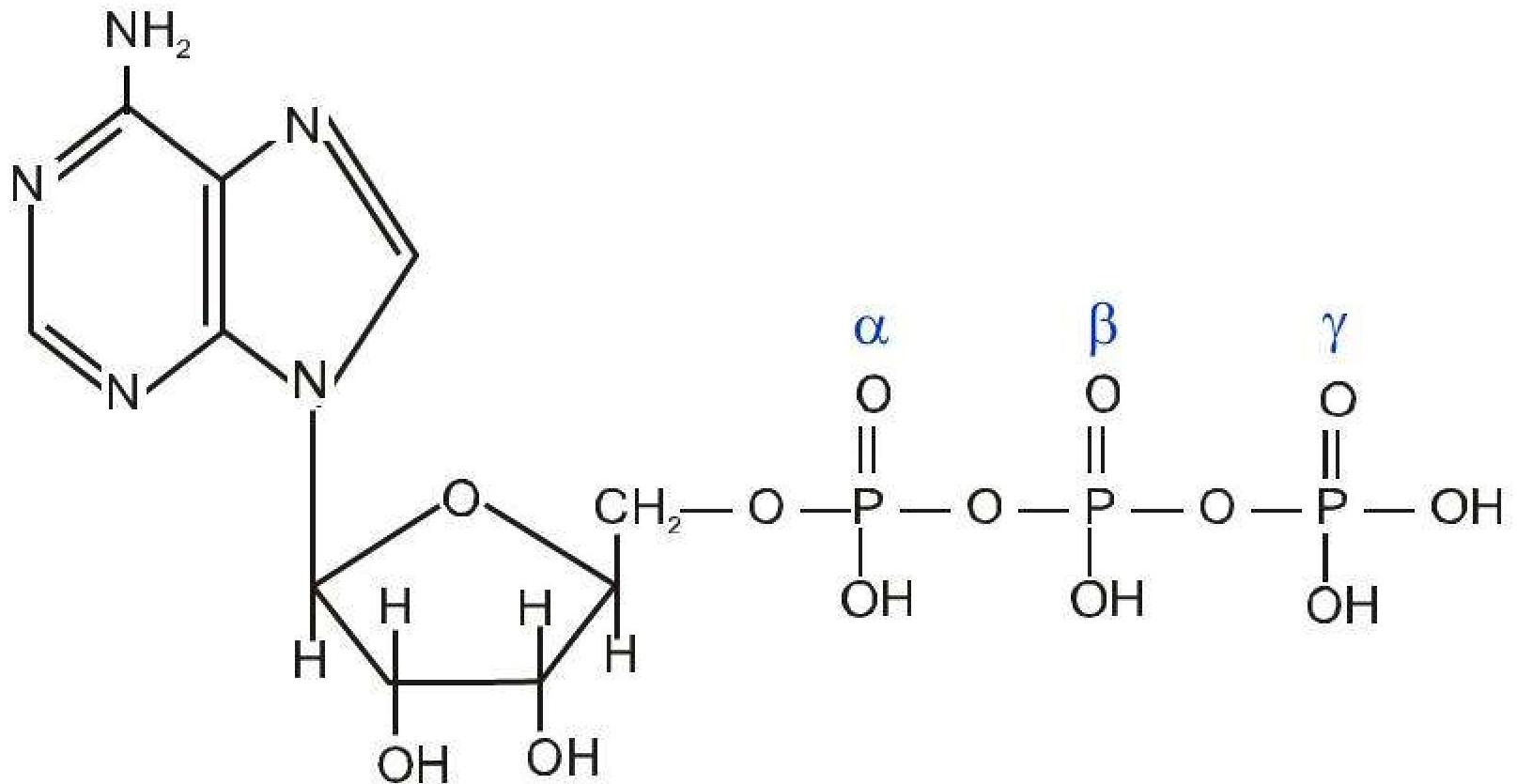
Removal of last two phosphates of ATP liberates a large amount of energy

Removal of the last phosphate of ADP liberates a large amount of energy

Therefore, ADP and ATP are said to be high-energy phosphates

ATP and ADP can be used as
source of energy in
energy-consuming reactions

During biological oxidation, energy is
captured in the form of ATP by
phosphorylation of ADP

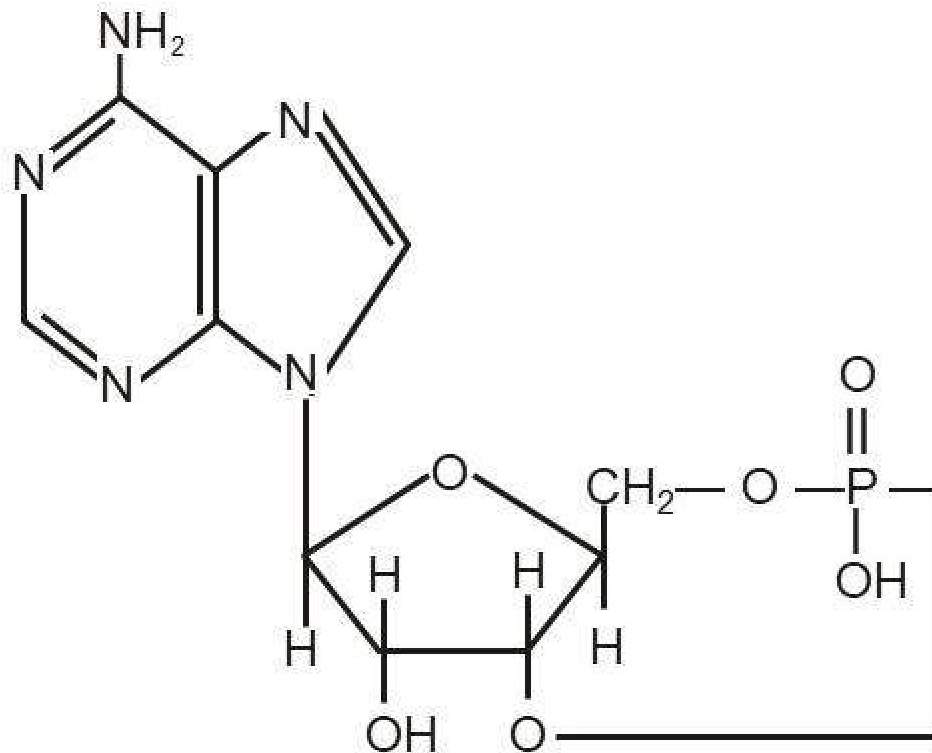


Adenosine-5'-monophosphate (AMP)

Adenosine-5'-diphosphate (ADP)

Adenosine-5'-triphosphate (ATP)

Another nucleotide derived from adenine is adenosine-3', 5'-monophosphate or cyclic AMP (cAMP)



It mediates the actions of several hormones

Cyclic AMP is formed by the action of adenylate cyclase on ATP

It can be converted into AMP by another enzyme, cAMP phosphodiesterase



Guanine derivatives

The guanine-containing nucleotides are GMP, GDP, GTP and cyclic GMP

GDP and GTP are high-energy phosphates

Cyclic GMP is a second messenger, and is also involved in the process of vision



Cyclic GMP is formed by the action of guanylate cyclase on GTP

Cyclic GMP can be converted into GMP by cGMP phosphodiesterase

Hypoxanthine derivatives

Hypoxanthine and ribose form a nucleoside, inosine

Inosine monophosphate (IMP) is the nucleotide formed from inosine

IMP is an intermediate in the synthesis of purine nucleotides

Cytosine derivatives

CMP, CDP and CTP are the nucleotides formed from cytosine

CDP and CTP are high-energy phosphates

CDP-Diacylglycerol and CDP-choline are intermediates in some pathways of lipid metabolism

Uracil derivatives

UMP, UDP and UTP are the uracil-containing nucleotides

UDP and UTP are high-energy phosphates

UDP-Glucose, UDP-galactose and UDP-glucuronic acid are intermediates in some metabolic pathways

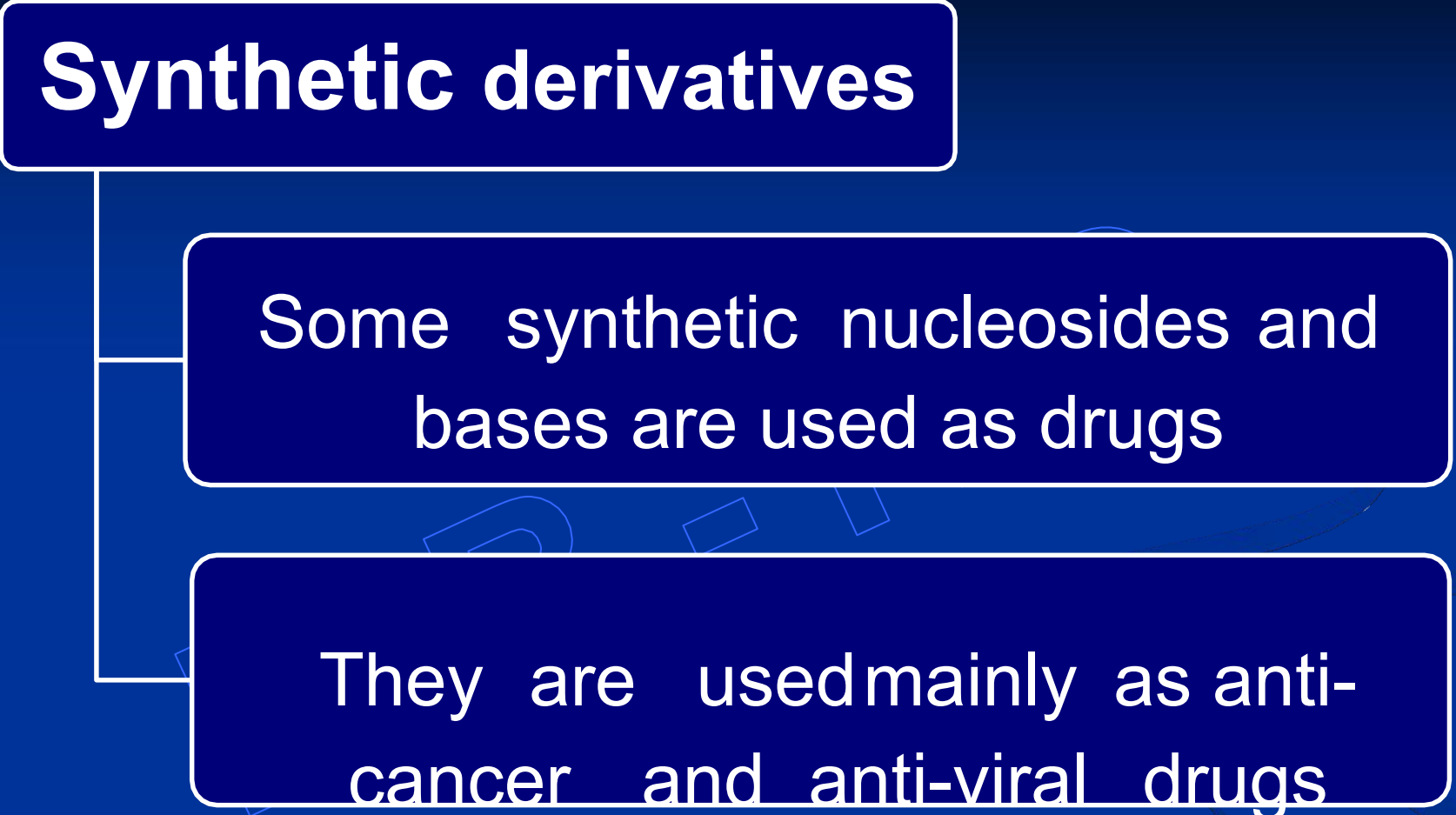
Vitamin derivatives

Most water-soluble vitamins perform their functions in the form of coenzymes

Many coenzymes are nucleotides containing adenine, ribose, phosphate and some other groups

Examples are flavin adenine dinucleotide (FAD), nicotinamide adenine dinucleotide (NAD) etc

Synthetic derivatives



```
graph TD; A[Synthetic derivatives] --- B[Some synthetic nucleosides and bases are used as drugs]; A --- C[They are used mainly as anti-cancer and anti-viral drugs]
```

Some synthetic nucleosides and bases are used as drugs

They are used mainly as anti-cancer and anti-viral drugs

Cell division requires replication of cellular DNA

Replication of DNA requires a ready supply of nucleotides

Abnormal nucleotides can block replication and cell division

Substitutions can be made in the structures of purine and pyrimidine bases

Substitutions can also be made in the sugars found in nucleosides

Substituted bases or nucleosides having substituted bases/sugars can be used as drugs

If substituted bases or nucleosides are given to a subject, they are incorporated in the nucleic acids

DNA having abnormal bases or sugars cannot be replicated, and this prevents cell division

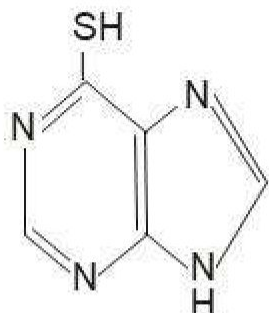
Some substituted bases/nucleosides are used as anti-cancer drugs

Some are used as antiviral drugs to check the multiplication of viruses

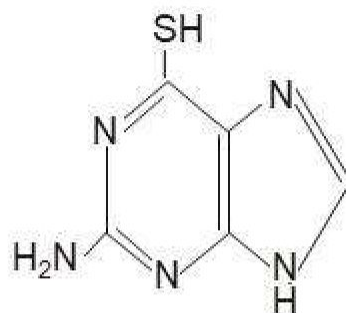
Examples of synthetic bases and nucleosides used as anti-cancer drugs:

- 6-Mercaptopurine
- 6-Thioguanine
- 5-Fluorouracil
- 5-Iododeoxyuridine
- Cytosine arabinoside
- Adenine arabinoside

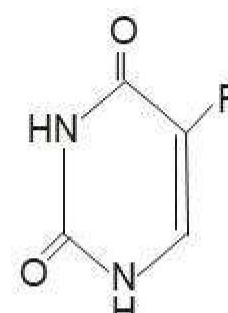
Synthetic bases and nucleosides used as anti-cancer drugs



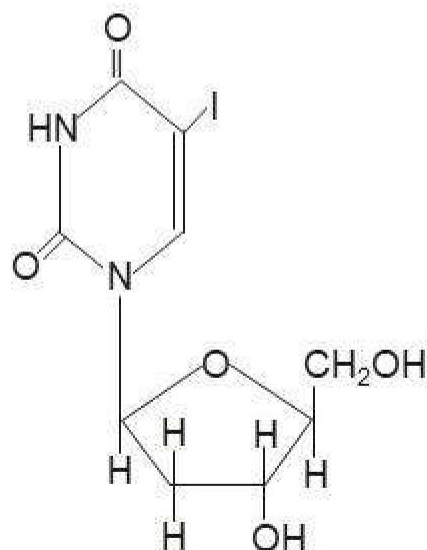
6-Mercaptopurine



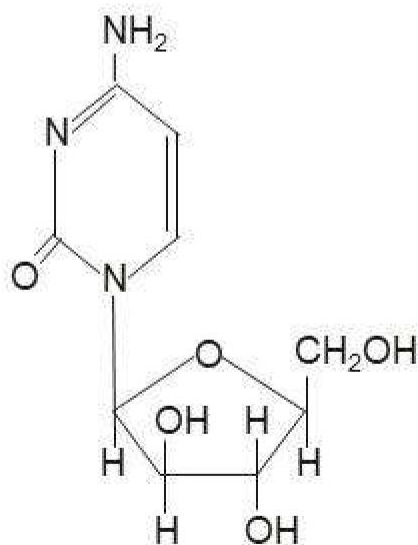
6-Thioguanine



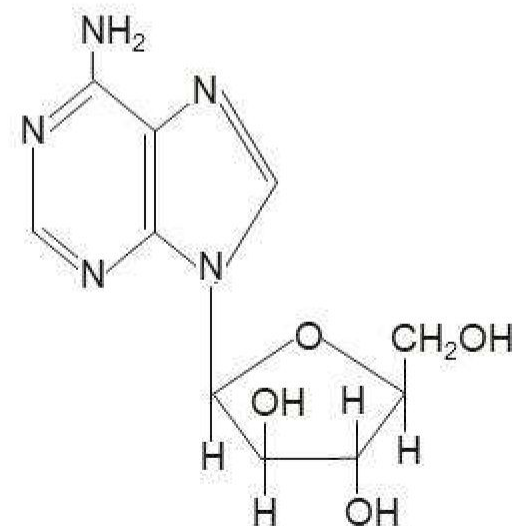
5-Fluorouracil



5-Iododeoxyuridine

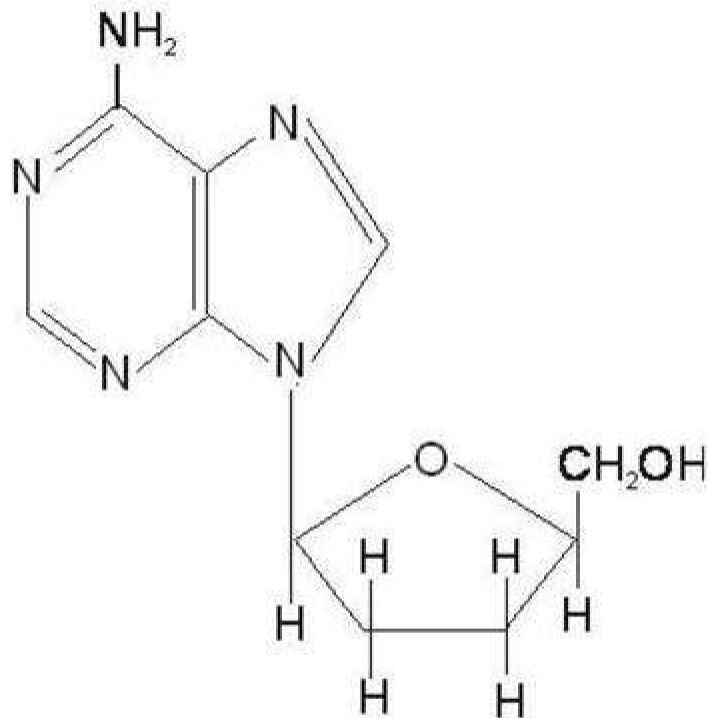


Cytosine arabinoside

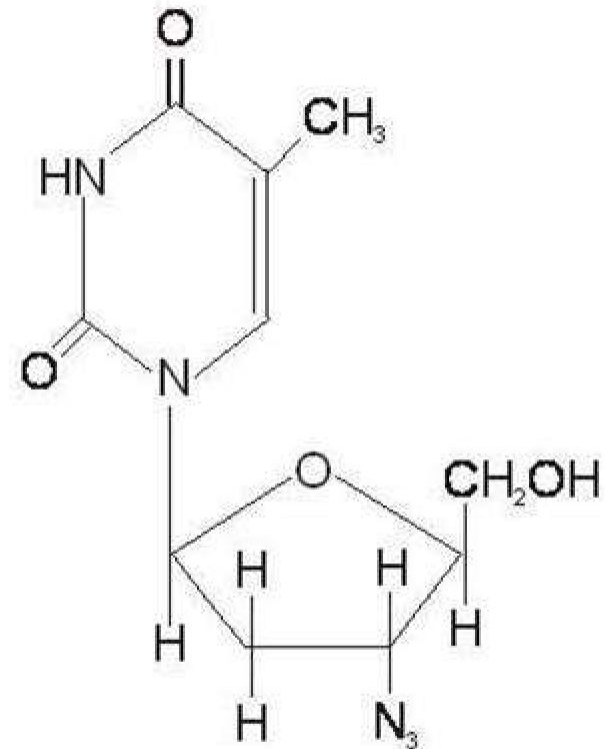


Adenine arabinoside

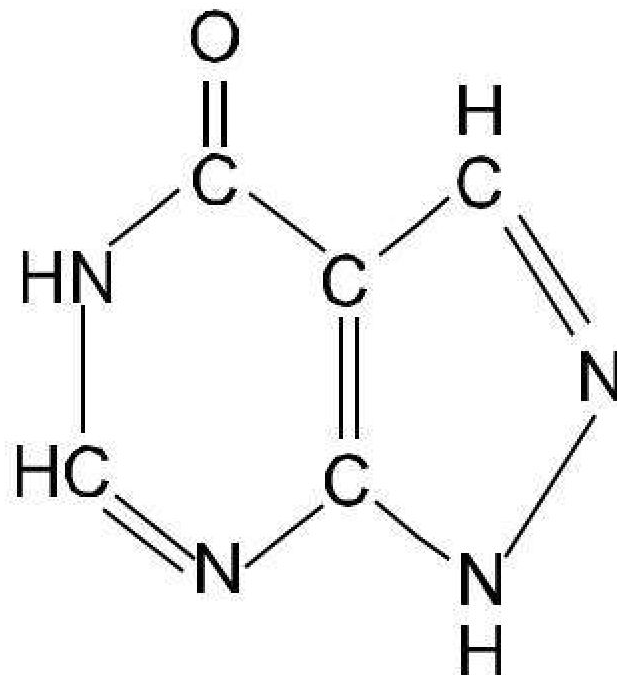
Synthetic nucleosides used in the treatment of AIDS



Dideoxyadenosine



Azidothymidine (AZT)



Allopurinol - A structural analogue of hypoxanthine used in the treatment of gout

Lippincott
Chatterjea
Harper

Abstract wavy lines in the bottom right corner of the slide, rendered in a lighter shade of blue than the background.